

TITLE PAGE

Problem Statement ID <<FILL IN>>

Problem Statement Title Eco-Loop Building Agents — autonomous closed-loop HVAC control using EnergyPlus, MCP and a local open-source LLM

Theme <<FILL IN>> · PS Category Software

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TECH STACK

EnergyPlus 26.1 · pyenergyplus · FastMCP · Model Context Protocol · Ollama + Llama 3.2 3B · Pydantic · Streamlit

A BUILDING THAT RUNS ITSELF

EnergyPlus supplies the physics. A local open-source LLM supplies the judgement. The Model Context Protocol carries every reading and every command between them — while the simulation is still running.

–8.5%

HVAC energy
summer week

336/336

agent turns
zero failures

100% open source

One laptop — no cloud, no API keys, no per-query cost

IDEA TITLE

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PROPOSED SOLUTION

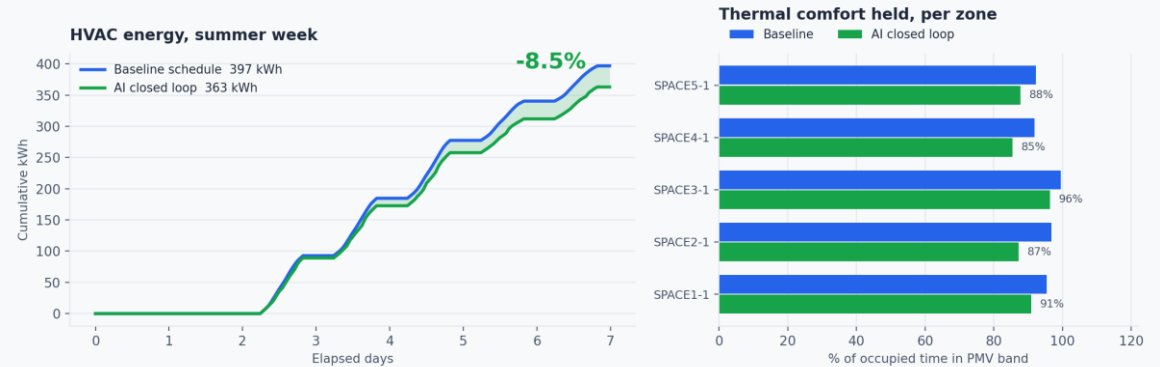
Traditional BMS follow rigid clock schedules. Eco-Loop turns the building into a self-correcting agent.

HOW IT WORKS

- EnergyPlus runs in-process; sensors read every zone timestep
- Every 60 simulated minutes the state reaches a local LLM via six MCP tools
- The model weighs comfort, demand and grid carbon, then returns setpoints

INNOVATION

- Injected into the LIVE solver — no IDF rewrite, no restart
- MCP is the safety boundary: every command validated server-side



–8.5% HVAC energy

Measured, not projected — while holding occupants closer to thermal neutrality than the baseline (mean PMV $-0.43 \rightarrow -0.02$)

WHY IT IS DIFFERENT

Most LLM-and-simulation work rewrites the model file and re-runs it — that can only compare finished runs. It cannot react to a zone drifting out of comfort at 14:00 on day 3.

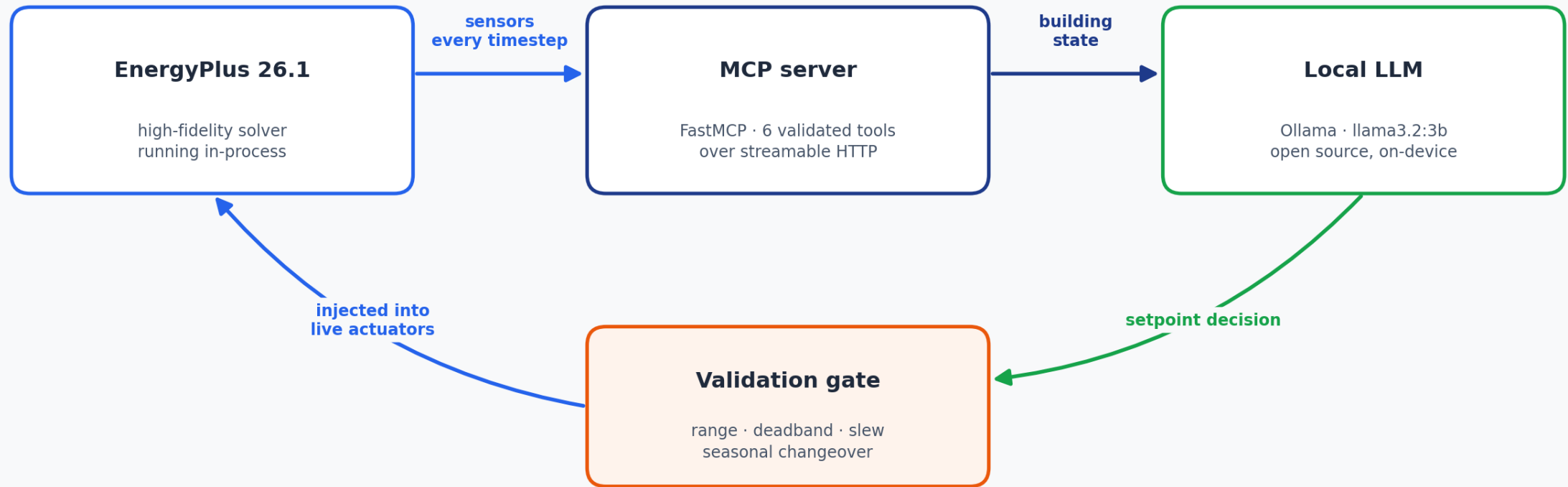
TECHNICAL APPROACH

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TECHNOLOGIES EnergyPlus 26.1 (pyenergyplus C API) · Python · FastMCP over streamable HTTP · Ollama running Llama 3.2 3B · Pydantic · Streamlit + Plotly

METHODOLOGY sample every timestep → aggregate per control interval → LLM tool-calling turn → validate → inject into live actuators

CLOSED LOOP · no file rewrite, no restart · 168 agent decisions per simulated week



The loop closes inside a single running simulation — 168 agent decisions per simulated week, zero restarts.

FEASIBILITY AND VIABILITY

336 / 336

agent turns — 0 fallbacks, 0 timeouts, 0 actuation errors

−8.5%

HVAC energy, summer week (−3.7% winter)

2.1 s

median agent decision latency, fully on-device

RISKS & CONSTRAINTS

- LLM latency inside a blocking simulation callback
- A small model can move the wrong setpoint, or run heating against cooling
- One setpoint pair cannot satisfy five differently-loaded zones

HOW EACH IS SOLVED

- Hard per-turn deadline; on timeout the loop holds the last accepted command
- Observations state the required direction — the model cannot invert physics
- Seasonal changeover parks the idle setpoint; a per-zone trim corrects each zone

VIABILITY AT SCALE

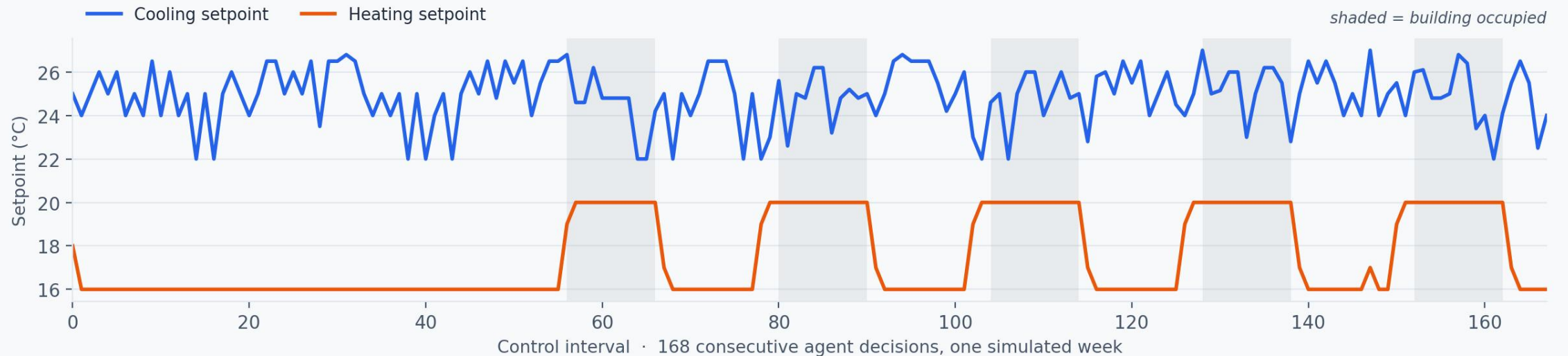
Buildings consume roughly 40% of global energy — a single-digit HVAC reduction is material at portfolio scale. The loop is model-agnostic: any EnergyPlus model with a thermostat works, and swapping in a larger cognitive engine needs no change to the MCP tool layer.

ARTIFACTS

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github.com/ayushYadav1107/Eco_loop

Baseline and runtime-generated .idf models, the full agent decision trail (agent_decisions.jsonl), per-timestep EnergyPlus results and the savings dashboard are all committed — a fresh clone reproduces every figure here.



The agent's own stated reasoning, taken verbatim from agent_decisions.jsonl

- 07-17 09:00** *"Raise cooling_sp to about 27.0 C this turn — this fixes comfort AND saves energy"*
- 07-17 11:00** *"HOLD the current setpoints. Do not trim energy here — you would push occupants out of the band"*
- 07-17 12:00** *"Nudge cooling_sp up to save energy while keeping PMV within the band"*

RESEARCH AND REFERENCES

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SIMULATION & BUILDING PHYSICS

- EnergyPlus Engineering Reference · EMS Application Guide
- pyenergyplus Data Exchange API — `get_variable_handle`, `set_actuator_value`
- ISO 7730:2005 · ASHRAE Standard 55 — PMV/PPD comfort model

AGENT, PROTOCOL & INFERENCE

- Model Context Protocol — `modelcontextprotocol.io` · FastMCP — `gofastmcp.com`
- Meta Llama 3.2 served locally through Ollama — `ollama.com`

REPRODUCE THE RESULT

```
python main.py prepare
python main.py run-baseline --start 07-15 --end 07-21
python main.py run-ai --start 07-15 --end 07-21
python main.py dashboard
```

AUDITABLE, NOT ANECDOTAL

The controller runs at temperature 0, so repeated runs return identical totals. Anyone can clone the repo and reproduce these exact figures.

HONEST LIMITATION — AND THE NEXT STEP

One setpoint pair drives five differently-loaded zones. Per zone the agent holds 85–96% of occupied time inside the PMV band; the stricter worst-zone metric is lower. Per-zone setpoints are next — the actuator handles and per-zone PMV are already in place.